

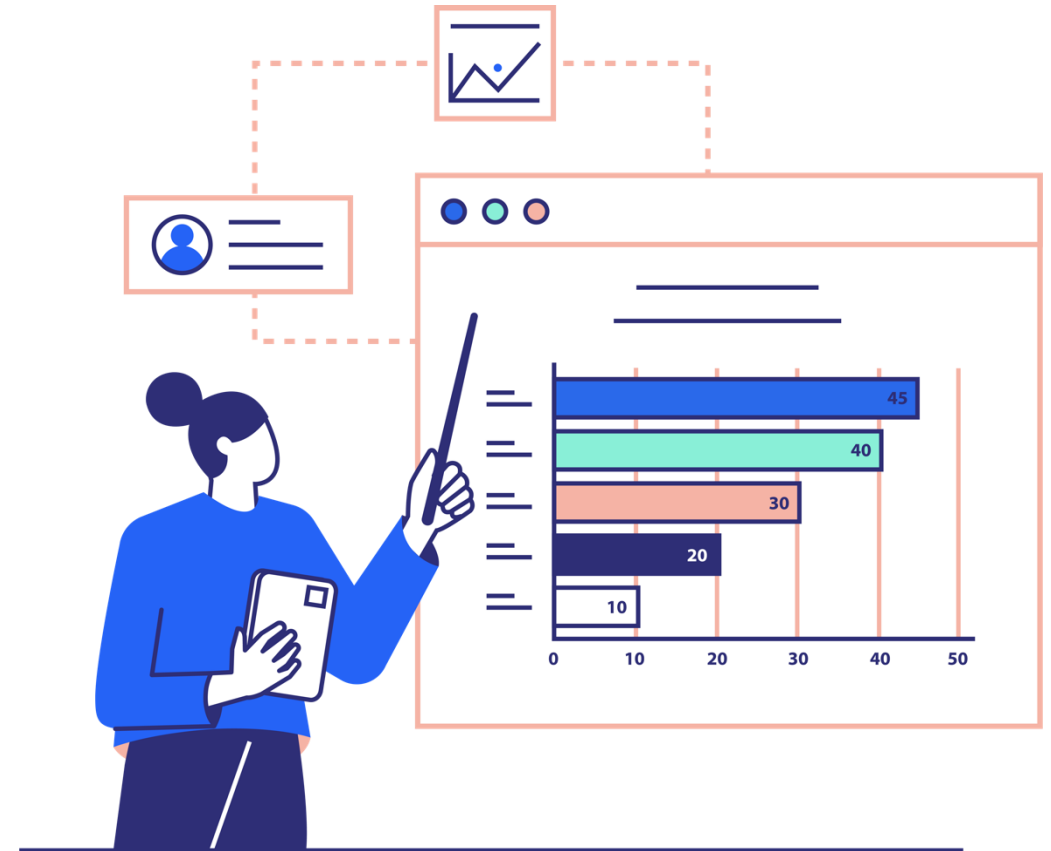
BEPEC Presentation

Advanced Statistics

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What is **Statistics**?

- ❑ Statistics is a branch of mathematics that deals with the collection, analysis, interpretation, and presentation of data. It helps in making data-driven decisions, identifying patterns, and drawing conclusions based on evidence.
- ❑ Statistics is broadly classified into **Descriptive Statistics** and **Inferential Statistics**.



Descriptive **Statistics?**

❑ Descriptive statistics summarize and describe the main features of a dataset. They do not make predictions but help in understanding the characteristics of the data.

➤ **Measures of Central Tendency (Show the center of data)**

1. Mean (Average): Sum of all values divided by the total number.
2. Median: The middle value when data is arranged in ascending order.
3. Mode: The most frequently occurring value.

➤ **Measures of Dispersion (Show the spread of data)**

1. Range: The difference between the maximum and minimum values.
2. Variance (σ^2): The average squared deviation from the mean.

$$\sigma^2 = \frac{1}{n} \sum (x_i - \mu)^2$$

3. Standard Deviation (σ): The square root of variance, indicating spread.
4. Interquartile Range (IQR): Difference between the 75th percentile (Q3) and the 25th percentile (Q1).

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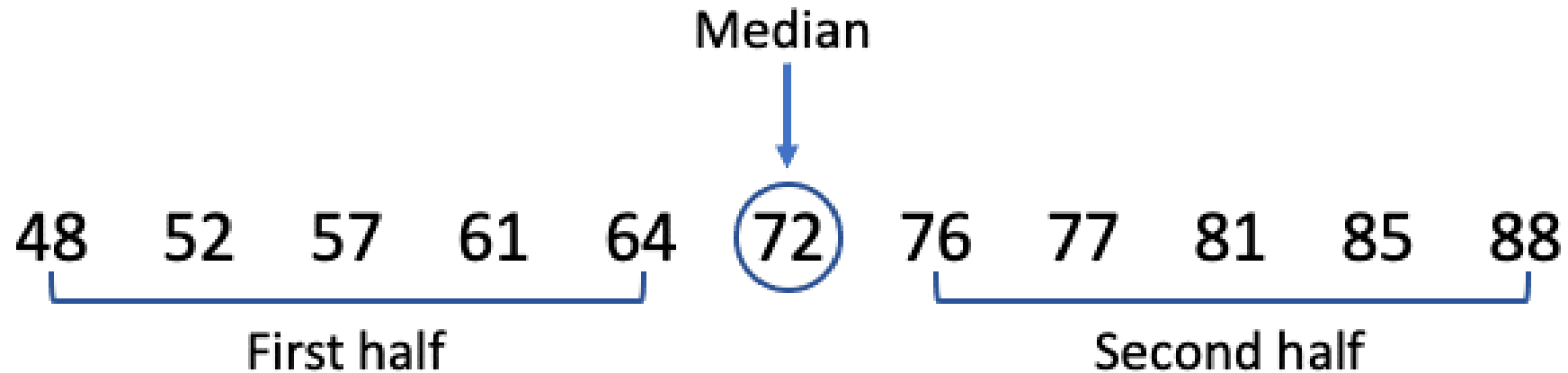
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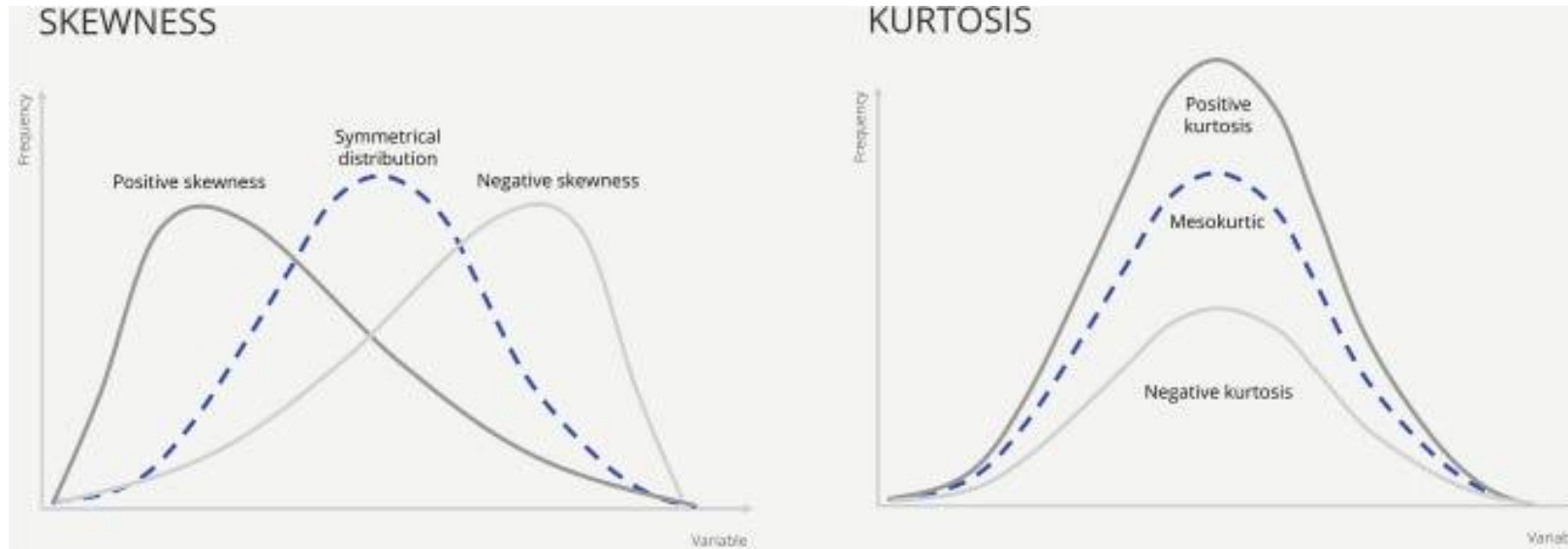
Descriptive **Statistics?**



Descriptive **Statistics?**

➤ Measures of Shape and Distribution

1. **Skewness:** Measures the asymmetry of the data distribution.
2. **Kurtosis:** Measures the "peakedness" or "flatness" of a distribution.



Descriptive **Statistics?**

➤ Data Visualization Techniques

1. Histogram
2. Box Plot
3. Scatter Plot
4. Bar Chart
5. Pie Chart.

Exploratory **Data Analysis?**

- ❑ EDA is primarily a part of **Descriptive Statistics**, as it focuses on summarizing and understanding data rather than making predictions. It helps in:
 1. Summarize the Data using Mean, Median, Variance
 2. Understanding Data Distributions using Skewness, histograms
 3. Detecting Outliers using IQR, Boxplots
 4. Finding Relationships between variable using
 1. Uni-variate
 2. Bi-Variate
 3. Multi-Variate
 5. Handling Missing Values using Imputation Techniques like mean, median, mode

What is **Inferential Statistics**?

Inferential Statistics is a branch of statistics which deals with making predictions or conclusions about population based on sample

What is **Probability**?

Probability is a measure of the likelihood that an event will occur. It quantifies uncertainty and is expressed as a number between 0 and 1

- ❑ **An experiment** is a process that leads to one of several possible outcomes.
 - ❑ **Example:** Rolling a die, tossing a coin, drawing a card from a deck.
- ❑ **The sample space** is the set of all possible outcomes of an experiment.
 - ❑ Tossing a coin: $S = \{H, T\}$
 - ❑ Rolling a die: $S = \{1, 2, 3, 4, 5, 6\}$
- ❑ **An event** is a subset of the sample space, consisting of one or more outcomes.
 - ❑ Rolling an even number on a die: $A = \{2, 4, 6\}$
 - ❑ Getting heads in a coin toss: $A = \{H\}$

$$P(A) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}}$$

Types of Events?

- **Mutually Exclusive Events:** Events that cannot happen at the same time.
 - **Example:** Rolling a die and getting either 2 or 5. You cannot roll both at the same time.

$$P(A \cap B) = 0$$

- **Independent Events:** Events where the occurrence of one does not affect the occurrence of the other.
 - **Example:** Tossing a coin and rolling a die are independent events.

$$P(A \cap B) = P(A) \times P(B)$$

- **Dependent Events:** Events where the outcome of one event affects the probability of another.
 - **Example:** Drawing two cards from a deck **without replacement**.

$$P(A \cap B) = P(A) \times P(B|A)$$

Types of **Events**?

- **Conditional Probability:** The probability of event AA occurring given that event BB has already occurred:
 - **Example:** Probability of drawing a king given the first card drawn was a king.

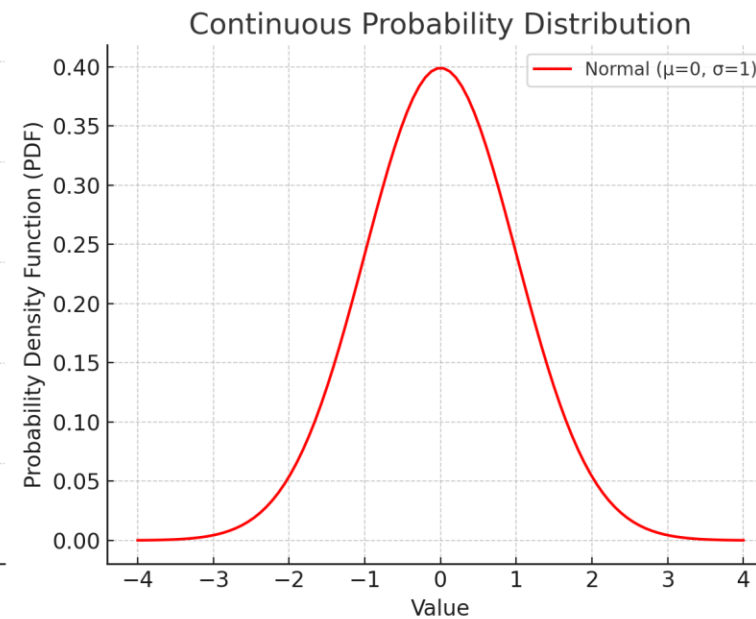
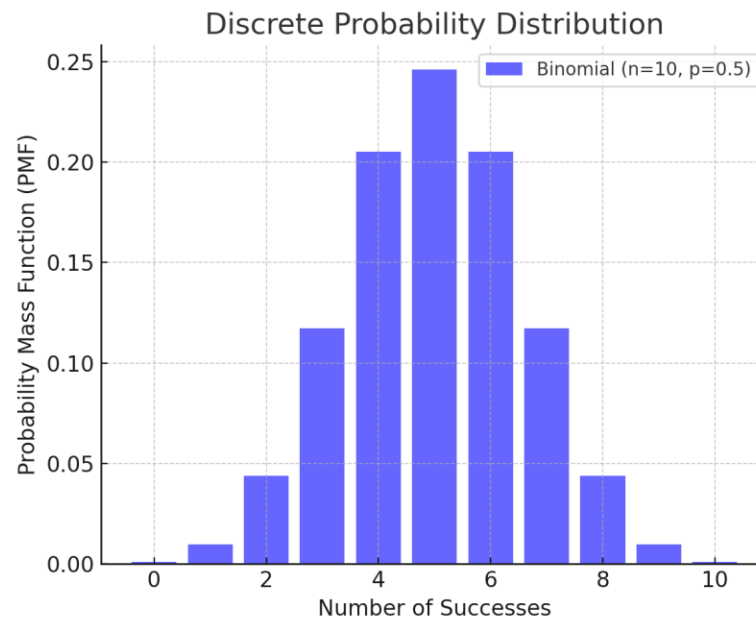
$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

- **Bayes' Theorem:** Used for updating probabilities based on new evidence
 - **Example:** Diagnosing a disease given a positive test result.

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Probability Distributions

- Defines how probabilities are distributed over possible values of a random variable.
 - **Discrete Distributions:** Bernoulli, Binomial, Poisson
 - **Continuous Distributions:** Normal, Exponential, Uniform



Discrete Probability Distributions

- A **discrete probability distribution** deals with **countable** outcomes (e.g., rolling a die, flipping a coin).
 - **Probability Mass Function (PMF)**: The **PMF** gives the probability of a discrete random variable taking a specific value.

$$P(X = x) = \frac{1}{n}, \quad \text{for } x = 1, 2, 3, 4, 5, 6$$

$$P(X = x) = \frac{1}{6} = 0.1667$$

Discrete Probability Distributions

- The **CDF** is the probability that X is less than or equal to a given value.

$$F(X) = P(X \leq x) = \sum_{k=1}^x P(X = k)$$

$$P(X \leq 2) = P(1) + P(2) = 0.1667 + 0.1667 = 0.3333$$

$$P(X \leq 4) = 0.1667 + 0.1667 + 0.1667 + 0.1667 = 0.6667$$

Continuous Probability **Distributions**

➤ Let's consider a **normal distribution** with:

❑ Mean (μ) = 0

❑ Standard deviation (σ) = 1

❑ Probability Density Function (PDF)

For a normal distribution:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

For $X = 0$, using $\mu = 0, \sigma = 1$:

X	-2	-1	0	1	2
PDF	0.0540	0.2420	0.3989	0.2420	0.0540

$$\begin{aligned} f(0) &= \frac{1}{1 \times \sqrt{2\pi}} e^{-\frac{(0-0)^2}{2(1)^2}} \\ &= \frac{1}{\sqrt{2\pi}} e^0 = \frac{1}{\sqrt{2\pi}} = 0.3989 \end{aligned}$$

Continuous Probability **Distributions**

- The **CDF** is computed using the integral of the PDF:

$$F(X) = P(X \leq x) = \int_{-\infty}^x f(t)dt$$

X	-2	-1	0	1	2
CDF	0.0228	0.1587	0.5000	0.8413	0.9772

Law of Large **Numbers**

- The Law of Large Numbers (LLN) states that as the number of observations increases, the sample mean Converges to the true population mean μ :

$$\lim_{n \rightarrow \infty} \bar{X} = \mu$$

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

Central Limit **Theorem**:

- The **Central Limit Theorem (CLT)** states that the sampling distribution of the **sample mean** (or sum) of a large number of **independent, identically distributed (i.i.d.)** random variables **approaches a normal distribution**, regardless of the original population's distribution.

Confidence **Interval**:

- A **Confidence Interval (CI)** is a range of values that likely contains the **true population parameter** (e.g., mean, proportion) with a certain level of confidence.
- It is commonly used in inferential statistics to estimate **population parameters** based on a **sample**.

Confidence **Interval**:

- For a population **mean** μ , when the population standard deviation σ is **known**, the confidence interval is

$$CI = \bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

- When σ is **unknown**, we use the sample standard deviation s and the **t-distribution**

$$CI = \bar{X} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$$

Hypothesis **Testing**:

- Hypothesis testing is a **statistical method** used to make decisions or inferences about a population **based on a sample**.
- It helps determine whether a claim (hypothesis) about a population parameter is **supported by sample data** or not.
 - **Null Hypothesis (H_0)**: It usually represents **no effect, no difference**
 - The average height of students is 170 cm ($H_0: \mu = 170$)
 - **Alternative Hypothesis (H_a)**: It represents a **difference, effect, or change**.
 - The average height of students is **not** 170 cm ($H_1: \mu \neq 170$)

Hypothesis **Testing**:

- **Test Statistic:** A numerical measure used to determine whether to reject H_0
 - It follows a specific probability distribution (e.g., Z-distribution, t-distribution, Chi-square, F-distribution)
- **P-Value:** The probability of obtaining the observed data (or something more extreme) assuming H_0 is true.
 - If **p-value** $< \alpha$, reject H_0 .
 - If **p-value** $\geq \alpha$, fail to reject H_0 .
 - **Type I Error** (α): Rejecting H_0 when it is actually true
 - **Type II Error** (β): Failing to reject H_0 when it is actually false

Steps in Hypothesis **Testing**:

- Step 1: Define Hypotheses
- Step 2: Select Significance Level
- Step 3: Choose a Test Statistic
- Step 4: Compute the Test Statistic
- Step 5: Calculate the P-Value
- Step 6: Compare P-Value with α
- Step 7: Draw a Conclusion

Steps in Hypothesis **Testing:**

Y	X	Test
Continuous	Discrete more than 2 categories	ANOVA (single factor)
Continuous	Discrete in 2 categories	t-Test
Continuous	Continuous	Regression
Discrete	Discrete in 2 categories	2-proportion test
Discrete	Discrete multiple categories	Chi-square test

Steps in Hypothesis **Testing:**

